

Automatic Hydroponic Plant Protection from rain

1. Introduction:

This simulation explores the behavior of an automated hydroponic plant cover system, which automatically closes during rainfall and reopens when the rain stops. The simulation focuses on verifying the activation of the DC gearbox when rain is detected by the sensor, and on ensuring that both limit switches correctly stop the motor when the cover reaches its fully closed or fully opened positions.

2. Simulation Setup and Design:

Falstad circuit simulator was used for the circuit simulation. The simulated circuit diagram closely followed the schematic from the project proposal, with a few adjustments to reflect the latest design changes based on the progress report.

Default Falstad components models were used for most of the components, except for the op-amp, which was configured to simulate the LM741 as used in the actual circuit. The relay coil resistance was set to 320Ω , based on the datasheet assumption that the coil consumes approximately 0.45 watts of power. A 12V DC voltage source was used to represent the 12V power adapter.

The rain sensor's behavior was simulated using a potentiometer configured as a rheostat. Its resistance was manually varied during the simulation to represent the change in conductivity of the rain sensor surface when wet and dry.

Simulations were performed to observe the circuit's switching behavior as the rain sensor resistance changed, representing the presence or absence of raindrops on the sensor surface.

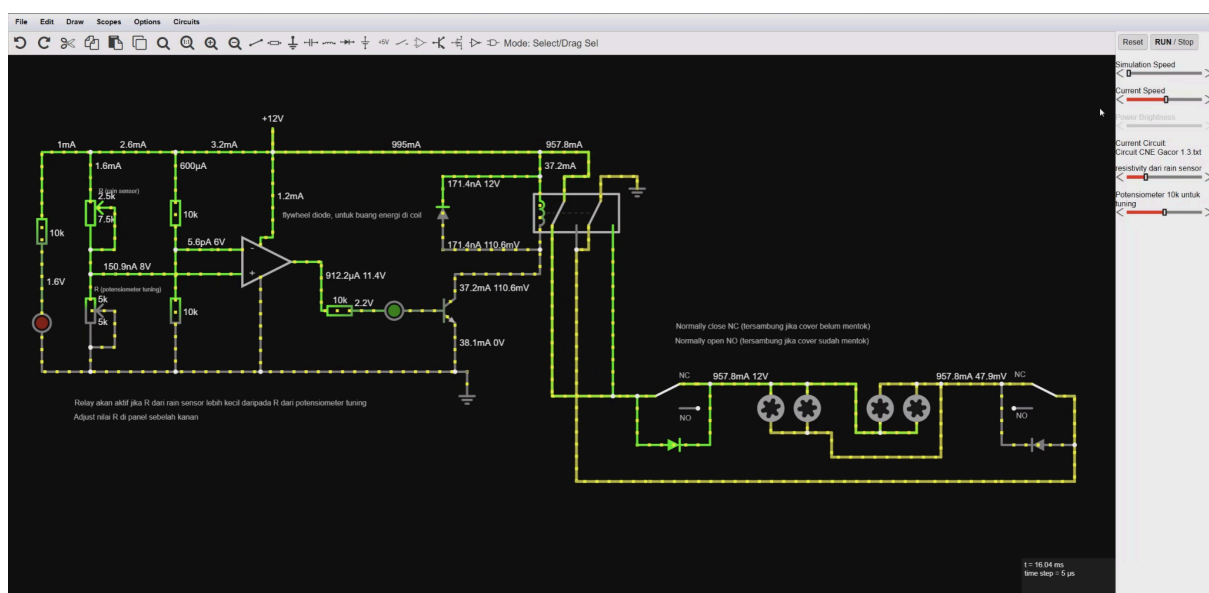


Figure 1: Simulation screenshot when rain is detected (sensor surface wet)

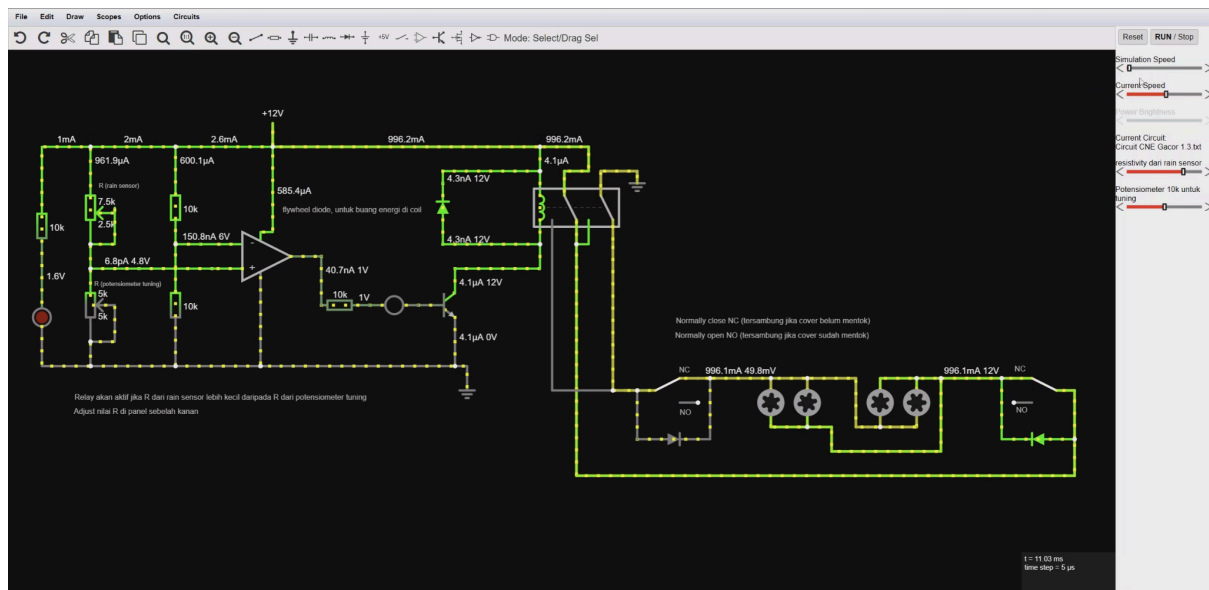


Figure 2: Simulation screenshot when rain stops and sensor surface dries

Component Models:

Default Falstad models were used for all components except as noted above. The rain sensor was simulated by using a potentiometer in rheostat mode, with manual adjustment of resistance values during the simulation to represent varying levels of wetness on the sensor surface.

Simulation Parameters:

The simulation was run without a predefined time duration. Instead, it was stopped when the total current drawn from the 12V supply approached the power adapter's current rating (1A). The rheostat resistance that simulates the rain sensor resistance was adjusted manually during the simulation to explore behavior at two main points: when the rain sensor resistance is lower than the tuning rheostat resistance, and when the rain sensor resistance is higher than the tuning rheostat resistance. These two points reflect the threshold behavior of the comparator circuit. Default Falstad simulation settings were used throughout to ensure consistency.

Test Circuits:

The primary simulation focused on the complete rain detection and motor control circuit. No separate test circuits were used for individual subsystems, as the focus was to evaluate the behavior of the fully integrated system.

3. Simulation Results and Analysis:

Presentation of Results:

The circuit was simulated using Falstad Circuit Simulator to test the full functionality of the rain detection and actuator control system. The simulation modeled the behavior of key components such as the LM741 comparator, BC547 transistor, relay, and N20 DC motor activation, based on rain sensor input.

The rain sensor was simulated using a potentiometer configured as a rheostat, allowing manual adjustment of resistance to represent wet and dry conditions. The switching behavior was observed in response to these changes.

Key Observations:

Condition	Rain Sensor Resistance	Op-Amp Output	Relay State	Motor Status
Dry (No Rain)	> 5kΩ	~0V	OFF	Cover open
Wet (Rain Detected)	< 5kΩ	~11.4V	ON	Cover closing
Rain Stops (Back to Dry)	> 5kΩ	~0V	OFF	Cover opening

Analysis and Interpretation:

The simulation results closely matched theoretical expectations. The comparator output voltage transitioned from low (~0V) to high (~11.4V) precisely when the rain sensor resistance dropped below the reference threshold, which is set by a voltage divider or tuning potentiometer (approximately 6V reference point).

This high output activated the BC547 transistor, which saturated and allowed current to flow through the relay coil. The relay then activated the motor circuit. When the rain stopped (resistance increased), the op-amp output returned to low, the transistor cut off, and the relay deactivated.

The motor behavior followed the relay state perfectly, with forward movement during rain and reverse movement when dry, controlled by the logic implemented using limit switches.

No unexpected behavior was observed. The transitions were clean, and no chattering or unstable switching occurred, indicating that the design has appropriate hysteresis and stable operation. Minor discrepancies in switching voltage occurred due to simulation tolerances, but these were within acceptable limits and do not affect real-world behavior.

Key Parameters:

Parameter	Simulated Value	Target/Theoretical Value	Notes
Comparator Switching Voltage	~6V (adjustable)	~6V	Controlled via 10k potentiometer

Op-Amp High Output	~11.4V	~12V max (supply-limited)	Sufficient to saturate transistor
Relay Coil Resistance	320Ω	Datasheet value	Confirmed 0.45W power draw
Relay Coil Current	~37.2mA	Theoretical: 37.5mA	Matches with 12V / 320Ω
Total Current Draw (Active)	~957.8mA	Limit: < 1A (adapter max)	Safe operating range

4. Comparison with Theoretical Calculations:

The calculated switching threshold for the comparator was based on the resistor divider formed by the rain sensor and the tuning potentiometer. At rain detection, the rain sensor resistance drops to around 2.5kΩ, producing approximately 8V at the non-inverting input of the LM741. The reference voltage was set to 6V using a 10kΩ resistor divider.

During simulation, the comparator switched as calculated when the rain sensor voltage goes above the reference voltage. The comparator output produced around 11.4V, which successfully drove the BC547 transistor into saturation, allowing approximately 37.2mA to flow through the relay coil. This activated the motor circuit, drawing a total current of about 957.8mA from the 12V power supply.

The simulation results are within a few percent of the calculated values, validating both the accuracy of the design calculations and the simulation model. Minor differences can be attributed to simulation simplifications, such as component tolerances and the simplified models used in the simulation.

5. Conclusion:

The simulation results confirmed that the circuit performs as expected based on the project's design goals. The comparator accurately responded to changes in the rain sensor resistance, switching the output at the appropriate threshold and correctly activating the relay and motor system. This behavior aligns with the intended function of automatically closing the hydroponic cover during rain and reopening it once conditions are dry.

Through the simulation process, we gained a deeper understanding of how component values such as resistor selection and reference voltage affect the circuit's sensitivity and reliability. We also identified areas for improvement, such as adding adjustable tuning via a potentiometer to fine-tune the switching threshold. These insights allowed us to refine the circuit before building the physical prototype, saving time and reducing troubleshooting later in the project.

Overall, the simulation validated the functionality of the entire system and provided confidence to proceed with PCB fabrication and full mechanical integration. It played a

crucial role in ensuring our design is both effective and efficient in achieving its purpose: protecting hydroponic plants from excess rain automatically.